

Manish Kumar

ENGINEERING LEADER · PRODUCT BUILDER

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Summary

A Computer Science and Engineering graduate from IIT Bombay with over 20 years of experience in systems and software development, specializing in Rust, C++, and CUDA. Proven expertise in building high-performance, GPU-accelerated and low-latency systems, including analytical engines and data platforms. Experienced in designing and delivering SaaS and desktop products end-to-end, and in hiring, mentoring, and leading engineering teams to build robust, scalable, and commercially successful products from scratch.

Skills

Programming Languages	Rust, C++, Python, Go, TypeScript
Databases	Amazon DynamoDB, Apache Cassandra, MongoDB, PostgreSQL, Redis, Scylla
Software Tools	AWS Lambda, Axum, CUDA, Docker, Django, GraphQL, gRPC, Tokio, Velox

Work Experience

OPEN SOURCE CONTRIBUTOR

- Active contributor to Apache Arrow Rust (arrow-rs) and Apache DataFusion, focusing on query execution and compute kernels.
- Building CUDA-based JIT-compiled analytics engine in Rust for GPU-accelerated expression evaluation ([tachyon](#)).
- Building SIMD-optimized K-Nearest Neighbors search engine in Rust ([nyas](#)).

DataPelago India Pvt. Ltd.

[Hyderabad, India](#)

CONSULTANT/MEMBER OF TECHNICAL STAFF

Apr. 2024 – Sep. 2025

- Contributed to the development of a GPU-accelerated database query execution engine for analytical workloads.
- Implemented CUDA-based kernels for mathematical evaluation, string processing, and type casting operations.
- Achieved over 4x performance improvement in expression evaluation compared to NVIDIA cuDF.
- Collaborated closely with query planner and execution teams to optimize end-to-end query latency.

NeoCraft Technologies Pvt. Ltd.

[Hyderabad, India](#)

CO-FOUNDER AND CTO

Oct. 2023 – Present

- Led architecture and development of high-performance backend systems for data-intensive and ML-driven products.
- Designed and delivered a recommendation engine using linear algebra and optimization techniques for a U.S.-based client.
- Migrated production workloads from AWS EC2 to AWS Lambda, reducing infrastructure costs by approximately 50 percent.
- Provided technical leadership, system design reviews, and hands-on implementation across multiple client engagements.

Turing Inc.

[Hyderabad, India \(Remote\)](#)

SENIOR SOFTWARE ENGINEER

Apr. 2020 – Sep. 2023

- Contributed to a large-scale supply chain management platform for a U.S.-based enterprise client.
- Designed and implemented backend features for a CRM system serving thousands of users.
- Improved system performance by approximately 50 percent through query optimization and backend refactoring.
- Worked across distributed teams, contributing to design discussions, code reviews, and production support.

Setuserv Info. Pvt. Ltd.

[Hyderabad, India](#)

PRINCIPAL SOFTWARE ENGINEER

Aug. 2018 – Mar. 2020

- Architected an NLP-based text analytics SaaS platform using a microservices architecture.
- Built and operated scalable data pipelines using Kafka, Cassandra, and ScyllaDB on AWS.
- Designed REST APIs for external consumers and gRPC-based internal service communication.
- Developed a high-dimensional analytics dashboard supporting millions of query combinations.
- Improved overall data processing throughput by 2x through architectural and algorithmic optimizations.
- Led and mentored a team of six engineers through design, implementation, and delivery.

Ebizon Net Info Pvt. Ltd.

[Noida, India](#)

CTO AND TECHNICAL ARCHITECT

Jul. 2015 – Jul. 2018

- Founded and scaled PushFlew, a customer lifecycle management SaaS platform for small and medium businesses.
- Built AppifyCart, a SaaS solution for creating and managing mobile commerce applications.
- Designed the backend architecture for Mofluid, a Magento-based mobile commerce platform.
- Owned system architecture, hiring, mentoring, and delivery for multiple concurrent products.

Geometric Ltd.

TECHNICAL SPECIALIST

- Developed a real-time 3D cloth simulation engine using NVIDIA CUDA for physics-based modeling.
- Implemented a geometric anomaly detection library for large-scale 3D engineering models.
- Led a team of five engineers across GPU computing and graphics-intensive projects.

Pune, India

Apr. 2010 – Jul. 2014

CSC India Pvt. Ltd. (PayPal IDC)

SOFTWARE ENGINEER

- Developed near real-time systems for fraud risk mitigation in PayPal payment workflows.
- Built a real-time address normalization and validation engine to improve data consistency.

Chennai, India

Oct. 2007 – Mar. 2010

Geometric Ltd.

SOFTWARE ENGINEER

- Developed surface modeling libraries for automated Class-A surface generation.
- Implemented optimization modules using Quadratic Programming techniques.

Pune, India

Jan. 2005 – Sep. 2007

IIT Bombay

JUNIOR RESEARCH FELLOW

- Researched optimization algorithms for automated Class-A surface design.
- Implemented Quadratic Programming models for CAGD applications using MATLAB.

Mumbai, India

Aug. 2004 – Jan. 2005

Education

IIT Bombay(Indian Institute of Technology Bombay)

B.TECH. IN COMPUTER SCIENCE AND ENGINEERING

Mumbai, India

Jul. 2000 - Aug. 2004